A Survey on Sim-to-Real Transfer Methods for Robotic Manipulation

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***Abstract*—The effective deployment of robotic systems in real-world environments requires the development of reliable control policies, which can be challenging due to safety, cost, and time constraints associated with direct real-world training. Sim-to-real transfer provides a solution by allowing robots to learn policies in simulated environments that can be seamlessly applied to real-world scenarios. This comprehensive review examines key methodologies for sim-to-real transfer in robotic manipulation, including domain randomization, domain adaptation, meta-learning, knowledge distillation, policy performance correlation, "green-screening" backgrounds, alternative 3d representations, and video generation. These approaches are critical for bridging the gap between simulated and actual environments, thereby facilitating the safe and efficient deployment of robotic systems. The review categorizes recent significant studies, explores major application areas and identifies primary challenges and future research directions. Our objective is to provide a thorough overview of current progress and to highlight promising avenues for future research in sim-to-real transfer, ultimately enhancing the robustness, efficiency, and adaptability of robotic systems in diverse real-world applications.**

***Index Terms*—Sim-to-Real Transfer, Robotic Manipulation, Domain Randomization, Deep Reinforcement Learning, Real-World Deployment**

I. Introduction

Deploying robotic systems in real-world environments necessitates robust control policies. Direct training in real settings poses challenges due to safety, cost, and time constraints. Sim-to-real transfer addresses these issues by enabling robots to learn in simulated environments before applying their policies in the real world. This review covers key sim-to-real transfer methods, including domain randomization, domain adaptation, imitation learning, meta-learning, knowledge distillation, policy performance correlation, "green-screening" backgrounds, alternative 3d representations, and video generation.

Domain randomization (DR) exposes robots to varied scenarios within simulations, improving their ability to generalize to real-world conditions. This technique reduces the "reality gap" between simulated and actual environments by ensuring the robot's training includes diverse conditions.

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Domain adaptation minimizes differences between simulated and real environments, enhancing the robustness of trained policies. Techniques like adversarial training and fine-tuning align the feature spaces of source and target domains, ensuring adaptability in real-world tasks.

Imitation learning uses expert demonstrations to train robots, allowing them to mimic human actions and quickly acquire complex skills. This method leverages human expertise to enhance robotic performance in learning tasks.

Meta-learning, or "learning to learn", enables robots to adapt quickly to new tasks by leveraging prior experiences. By training on various tasks, robots develop a meta-policy for rapid adaptation to new, unseen tasks, making them versatile in dynamic environments.

Knowledge distillation transfers knowledge from complex models to smaller, efficient ones, allowing lightweight models to retain performance capabilities. This technique is beneficial for real-time applications on resource-constrained devices.

Recent research [1] shows that there is a significant correlation between how policies perform in simulation and in real-world scenarios, emphasizing the importance of simulators for evaluating robotic policies.

NeRF [2] and 3D-GS [3] are alternative methods for 3D scene reconstruction and real-time rendering, with recent research [4] enabling dynamic 3D robot representations at 30Hz using three cameras while synchronizing with reality.

Video generation approach [5] uses internal representations to create realistic simulations, enhancing visual fidelity and robot perception by mapping input data to a latent space.

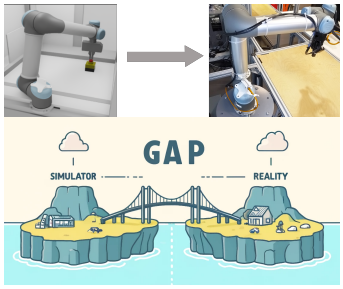


Fig. 1. Sim-to-real concept in robotic manipulation. It involves applying policies trained in simulation environments to real-world contexts.

Limitations include long training time, occasional object disappearance, and limited physical accuracy.

This review also highlights major applications such as dexterous manipulation and robotic navigation, and identifies primary challenges in sim-to-real transfer, including the reality gap and the need for high-fidelity simulations. By advancing these methodologies, we aim to improve the robustness, efficiency, and adaptability of robotic systems for diverse real-world applications (see Table I).

In summary, sim-to-real transfer is crucial for developing effective control policies for real-world robotics. Through detailed examination of methodologies, applications, and challenges, this review contributes to the ongoing advancement of sim-to-real transfer techniques, ensuring successful deployment of robots across various industries (see Figure 2).

II. Background

A. Deep Reinforcement Learning

RL algorithms learn from the rewards received based on actions taken in various scenarios, necessitating extensive experience. In real-world environments, gathering such experience can be difficult due to safety risks and time requirements. Deep RL algorithms, which use neural networks to approximate functions and policies, have demonstrated substantial success in simulated environments [6, 7]. However, these algorithms often encounter difficulties when applied directly to real-world tasks, primarily due to discrepancies between simulated and real environments. For instance, models might overfit to simulated dynamics, leading to poor performance when facing real-world variations.

B. Sim-to-Real Transfer

Sim-to-real transfer involves applying policies trained in simulation environments to real-world contexts. This process aims to bridge the gap between simulated and real environments by addressing differences in sensing and actuation. Effective sim-to-real transfer methods are essential for maximizing the benefits of simulation-based training, such as cost efficiency and safety, while ensuring that the policies perform well in real-world applications. Key challenges include dealing with the reality gap, which encompasses differences in physics, sensor noise, and environmental dynamics between simulation and reality.

C. Knowledge Distillation

Knowledge distillation involves training a smaller, more efficient model (student) to replicate the performance of a larger, more complex model (teacher). This method is valuable in DRL, where models often have high-dimensional inputs. The student model learns to predict outputs using only the data available in the real world, benefiting from the teacher model's rich knowledge without needing direct access to privileged information. This approach is particularly advantageous when deploying models on resource-constrained devices where computational efficiency is crucial.

D. Zero-shot Transfer

Zero-shot transfer refers to applying policies trained in simulation directly to real-world settings without additional training. This approach relies on the accuracy and realism of the simulation environment and the robustness of the trained policies. However, achieving successful zero-shot transfer can be challenging due to inherent differences between simulated and real environments.

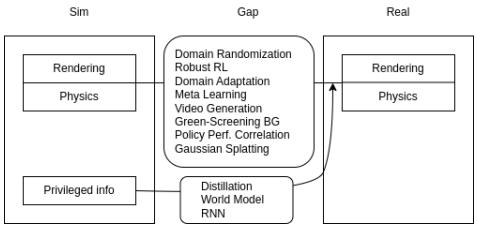


Fig. 2. Sim-to-real transfer techniques in robotic manipulation area.

III. Methodologies for Sim-to-Real Transfer

A. System Identification

System identification involves creating precise mathematical models of physical systems to improve the realism of simulations. This method requires careful calibration and measurement of system parameters. Challenges in system identification include variations in physical parameters due to environmental factors and wear over time, which can affect model accuracy. Techniques like residual physics learning and online adaptation help mitigate these issues by continuously updating model parameters based on real-world feedback [1].

B. Domain Randomization Methods

DR involves varying simulation parameters to create a diverse set of training scenarios. This helps models generalize to real-world variations. DR is often used to train policies for robotic manipulation, demonstrating its effectiveness in improving the robustness of trained models [8]. By exposing the model to a wide range of simulated variations, DR helps bridge the reality gap, making the model more resilient to unforeseen real-world scenarios [9, 10, 11, 12].

C. Transfer Learning and Domain Adaptation

Transfer learning seeks to improve the performance of models in target domains by leveraging knowledge from related source domains. Domain adaptation, a subset of transfer learning, focuses on scenarios where there is ample labeled data in the source domain but limited or no labeled data in the target domain. This technique is particularly useful in sim-to-real transfer, where the goal is to adapt policies learned in simulation to perform well in real-world settings with minimal additional training. Techniques like adversarial training and fine-tuning are commonly used to align the feature spaces of source and target domains, thereby enhancing the robustness of the transfer.

Domain adaptation aligns the feature spaces of source and target domains to facilitate knowledge transfer. Techniques such as adversarial training, where a discriminator is trained to distinguish between source and target domain data, and reconstruction-based methods, where a model is trained to reconstruct input data from both domains, are commonly used in domain adaptation. These methods aim to reduce the distribution shift between simulated and real data, thereby improving the model's performance in real-world tasks [13, 14, 15, 16, 17, 18].

D. Robust RL and Imitation Learning

Robust RL considers input disturbances and modeling errors to maximize reward. It aims to develop policies that are resilient to variations and uncertainties in the environment. Imitation learning, on the other hand, involves training models using expert demonstrations. This method provides robust rewards and can be particularly effective in sim-to-real transfer by leveraging human expertise to guide the learning process. Robust RL often incorporates techniques like noise injection and adversarial training to enhance policy robustness, while imitation learning employs techniques such as behavioral cloning and inverse RL.

Introducing perturbations in simulation environments can make agents more robust to real-world variations. This approach involves adding random forces or noisy rewards to better simulate real-world conditions. Learning with disturbances helps ensure that the policies trained in simulation can handle unexpected changes and uncertainties in the real world. Techniques such as robust adversarial RL and disturbance injection are commonly used to enhance the robustness of the learned policies [19].

E. Simulation Environments

Selecting the appropriate simulation environment is crucial for effective sim-to-real transfer. Common simulators like Gazebo, PyBullet, and MuJoCo each have unique strengths and integration capabilities. For instance, Gazebo is well-suited for complex scenarios and offers integration with the Robot Operating System (ROS), while PyBullet and MuJoCo provide faster training and are widely used in research. Custom simulations are often developed for specific tasks to meet particular requirements and constraints. The choice of simulator significantly impacts the success of sim-to-real transfer, as it determines the fidelity and realism of the training environment.

F. Meta Reinforcement Learning

MetaRL trains models to adapt quickly to new tasks by learning from various tasks. This approach typically incorporates memory mechanisms, such as long short-term memory networks, to retain and utilize past experiences. MetaRL aims to create models that can generalize across different tasks, making them highly adaptable to new and unforeseen scenarios. The objective is to develop a meta-policy that can quickly adapt to a new task with minimal additional training, leveraging the experience gained from related tasks [8].

G. Policy Performance Correlation

Simulators are essential tools not only for training robotic policies but also for evaluating their effectiveness. Recent research [1] shows that there is a significant correlation between how policies perform in simulation and in real-world scenarios. To bridge the control and visual gaps between simulated and real environments without needing full-fidelity digital twins, several techniques have been proposed: offline system identification, "green-screening" simulation observations using realworld backgrounds, and object texture baking from real-world images. In "green-screening" approach they overlay an image of the real-world environment onto the background of the simulated scene. Limitations include the use of static cameras only.

H. Video Generation

Some approaches use video generation to create realistic simulations by constructing internal representations. This involves mapping input data to a latent space, where high-level semantic information is extracted, allowing robots to perceive the essential aspects of the environment. IRASim approach [5] improves prediction and understanding of the simulated world, enabling realistic rendering. However, it also has limitations, including long training times (2369 hours on an A100 GPU), occasional disappearance of objects, operation primarily in 2D, and limited physical accuracy. Despite these challenges, the realistic rendering capability of IRASim enhances the visual fidelity of simulations.

I. Alternative 3D Scene Representations

Neural Radiance Fields (NeRF) [2] have become a prominent method for 3D scene reconstruction [20, 21, 22] and representation, utilizing input images as projective constraints to generate high-quality photorealistic views from a collection of RGB images. 3D Gaussian Splatting (3D-GS) [3] enhances the rendering process to real-time speeds by representing scenes with 3D Gaussian functions. These methodologies have been adapted to account for temporal dynamics. Recent research [4] has demonstrated the capability to create 3D representations of robots, incorporating realistic rendering and dynamic objects, operating in real-time at 30Hz with just three cameras.

IV. Application Scenarios

A. Dexterous Robotic Manipulation

Robotic manipulation tasks, such as peg-in-hole and deformable object manipulation, benefit significantly from sim-to-real transfer techniques [10]. For example, [16] used DR to train manipulation policies, addressing the challenges of simulating the deformability of objects. These tasks require precise control and adaptability, which can be enhanced through robust sim-to-real transfer methods. The ability to manipulate a wide range of objects with varying properties is critical for applications in industrial automation, healthcare, and domestic assistance.

TABLE I
ANALYSIS OF RELEVANT PUBLICATIONS ON SIM-TO-REAL TRANSFER FOR ROBOTIC MANIPULATION.

Paper	***Task***	***Input***	***Output***	***Motivation***	***Limits***	***Idea***
Jad Abou-Chakra et al. [4]	Deformables, pushover, pickup, rope, multiple objects, complex object	3 RGBD cameras	Control actions	Gaussian splatting provides fast and high-quality rendering, but does not work with dynamic objects.	Fails, when an object is moved very quickly by the robot. Planning is correct for several seconds only (4 sec.).	Generate "visual forces" that correct the particle positions while respecting known physical constraints. Add continuous visually-derived corrections.
Fangqi Zhu et al. [5]	Pushing	Train: video datasets. Inference: initial frame and 2D trajectory.	Extremely realistic videos of a robot arm that executes a given action trajectory.	To use the power of generative models for robotic arm control.	Long training time (2369 hours on an A100 GPU), occasional disappearance of objects, operation primarily in 2D, and limited physical accuracy.	Train generative model on huge amount of manipulation videos, mapping input data to a latent space, where high-level semantic information is extracted.
Xuanlin Li et al. [1]	Pick and place	Visual	Control actions	It is difficult to close the sim-to-real gap completely. Achieving correlation of policy effectiveness is a simpler task.	Using static cameras only.	Close correlation gap using: offline system identification, "green-screening" using realworld backgrounds, and object texture baking.
Garcia et al. [9]	Stacking, box-retrieving, assembling, pushing, pushing-to-pick, sweeping and ropeshaping	RGB images (3 frames history), gripper pos/rot	Gripper velocity, open/close command	Real-robot sim2real validation requires a significant amount of expensive iterations.	Fails, when during pushing the cube rotates by 90 degrees, a state never observed in training. Focused only on reducing the visual sim-to-real gap.	Pretrain sim2real on object localization task before manipulation.
Yuqing Du et al. [10]	Rope Peg-in-Hole and Cabinet Slide	Raw RGB	Control actions	Domain randomization requires extra knowledge to find how much to randomize params.	It is difficult to learn the true simulation parameters when the initial randomization is too far from the true values	It is hard to predict door damping, handle friction from watching door opening, but we can answer if damping in sim is stronger/weaker. Search Param Model (SPM) automatically tunes params.
Scheikl et al. [16]	Deformable object manipulation (cancer surgery)	Encoded RGB	Joints velocities	Visual sim-to-real gap was not bridged in surgery.	Camera calibration has huge effect.	Encode sim image (to be like real) using DCL (type of GANs). Train policy on encoded images from sim.
Zhang et al. [17]	Reaching a target in a changing environment, motion planning	RGBD	3d bounding box	Agnostic to the geometry and appearance representation of obstacles and target.	Bbox shape is "square", so it cannot handle complex shapes. Can't be used with lidar.	Vision-based actor-critic framework estimates the 3D bounding boxes of obstacles and targets from the RGB-D input.
Huy Ha et al. [23]	Transport task with five novel objects	Task text	Commands to robot	Applying LLM to manipulation	Dataset generation uses privileged info	Generate code for task. Generate code for success detector (retry if fail). 2 phases: generate data, distill data to policy (multi-task, visio-linguo-motor).
Liqian Ma et al. [24]	Manipulate an unseen articulated object in real world with no human demonstrations	2 point clouds	Trajectory	Solving long-horizon manipulation tasks	Low precision	Create 3d mesh with joints from 2 point clouds. Give this mesh to arm. Arm learns to move joints.
Juan et al. [25]	sim2sim: Pendulum, Walker2d, HalfCheetah; sim2real: Reach.	RGB camera on gripper	Gripper position and rotation	Sim2real dynamic transfer is poorly explored	Assumes state space is the same in sim and real	Train openai gym in sim1, test in sim2 with different gravity, friction. Use dynamics model, reward model, progressive NN for policy, 2 replay buffers.

B. Robotic Navigation

Sim-to-real transfer for robotic navigation encompasses tasks like mapless navigation, obstacle avoidance, and autonomous exploration. Techniques such as curriculum learning, where the complexity of tasks is gradually increased, and incremental environment complexity have shown promise in transferring navigation policies from simulation to reality. These methods help robots navigate unfamiliar and dynamic environments effectively. Applications in this domain include autonomous vehicles, drones, and service robots operating in dynamic and unpredictable environments [26].

C. Other Applications

Sim-to-real transfer has a wide range of applications beyond manipulation and navigation. For instance, it can be used in tactile sensing, where robots need to interpret and respond to physical touch [27]. Multi-agent systems also benefit from sim-to-real transfer, enabling coordinated actions and communications in tasks such as search and rescue or industrial automation. Other emerging applications include agricultural robotics, where robots perform tasks like crop monitoring and harvesting, and environmental monitoring, where robots collect data in challenging and hazardous environments.

V. Transferring Privileged Information Using World Models, Distillation, and RNNs

A. World Models

World models create internal representations of the environment, predicting future states from sequences of actions. They leverage privileged information from simulations, such as exact physical properties, to train robust policies. When applied to real-world tasks, these models generalize well, even in the absence of detailed information. World models can incorporate elements like latent space representations and predictive modeling to enhance their ability to simulate and anticipate future states, thus improving the robustness of the learned policies [28, 24, 25].

B. Knowledge Distillation

Knowledge distillation transfers knowledge from a larger, complex model (teacher) to a smaller, efficient model (student) that relies on observable data. This process retains crucial information without needing privileged information during real-world deployment, making it ideal for sim-to-real transfer. The distilled model benefits from the teacher model's comprehensive understanding of the task, while being lightweight and efficient enough for real-time deployment on resource-constrained platforms [23].

C. Recurrent Neural Networks (RNNs)

RNNs process sequences of data, maintaining a hidden state that captures information from previous inputs [29]. They are used to predict privileged information from observable data, enabling real-time adaptation to real-world dynamics. This capability makes RNNs particularly useful in sim-to-real transfer for tasks requiring continuous adjustment based on temporal patterns. RNNs can model temporal dependencies and long-term correlations in the data, which are crucial for tasks involving sequential decision-making and real-time control [8, 30].

D. Integrative Approach

Combining world models, knowledge distillation, and RNNs offers a robust strategy for sim-to-real transfer. The world model uses simulation data to train comprehensive policies, knowledge distillation simplifies the model for real-world use, and RNNs dynamically infer hidden parameters during deployment. This integrative approach ensures robust policies that can adapt to real-world variations effectively. By leveraging the strengths of each technique, the integrative approach enhances the robustness, efficiency, and adaptability of the learned policies, making them suitable for a wide range of real-world applications.

VI. Evaluation Metrics and Benchmarking

Evaluating the performance and robustness of models and techniques in sim-to-real transfer for robotic manipulation is essential. This chapter outlines the key evaluation metrics and benchmarking strategies used to assess the effectiveness of sim-to-real transfer methodologies.

A. Evaluation Metrics

Simulated vs. Real-World Performance:

- Transfer Success Rate: measures the proportion of tasks completed in the real world relative to the simulation [31].
- Task Completion Time and Error Rate: quantifies efficiency and accuracy of robotic performance, vital in assembly systems for benchmarking [32].

Robustness and Generalization:

- Robustness to Variations: Evaluates the model's ability to handle real-world variations and disturbances that were not encountered during training [33].
- Generalization Capability: evaluates model performance on unseen tasks and environments beyond training [34].

Efficiency and Scalability:

- Computational Efficiency: measures the computational resources, such as processing time and memory usage, required for training and deploying the model [17].
- Scalability: evaluates the model's ability to adapt to more complex tasks or larger environments without significant performance loss.

B. Benchmarking Strategies

Standardized Benchmarks:

- Common Datasets and Tasks: using standardized datasets and tasks, such as those from OpenAI Gym, to ensure comparability of results.
- Benchmarking Platforms: platforms like the DARPA Robotics Challenge offer standardized tasks and metrics for evaluating sim-to-real transfer techniques.

Real-World Testing Protocols:

- Controlled Experiments: conducting experiments where specific variables are systematically varied to understand their impact on sim-to-real transfer techniques [35].
- Field Tests: deploying robots in real-world settings to evaluate their performance under natural conditions and operational constraints.

Cross-Domain Validation:

- Domain Adaptation Benchmarks: using benchmarks designed for domain adaptation to evaluate how well models transfer knowledge from sim to real-world settings [36].
- Multi-Task Benchmarks: assessing models across various tasks to understand their generalization capabilities and robustness.

C. Case Studies and Comparative Analysis

Comparative Studies:

- Comparison of Techniques: comparing different sim-to-real transfer techniques, such as DR versus domain adaptation, to highlight their strengths and weaknesses.
- Performance Trends: analyzing performance trends over time to find areas of improvement and future research needs.

Case Studies:

- Application-Specific Evaluations: detailed evaluations of specific applications, such as robotic grasping, navigation, and manipulation, to provide insights into practical challenges and solutions in sim-to-real transfer.
- Industry Applications: case studies from various industries demonstrating the real-world impact and benefits of sim-to-real transfer techniques.

Incorporating detailed evaluation metrics and benchmarking strategies into sim-to-real transfer research is essential for assessing the effectiveness and robustness of developed models. The ongoing development of comprehensive evaluation methods will play a critical role in advancing the field and unlocking the full potential of sim-to-real transfer techniques in various robotic applications.

VII. Main Challenges and Future Directions

Despite significant progress, sim-to-real transfer remains a complex and challenging field. Designing efficient simulations and randomization distributions is intricate, and many domain adaptation algorithms assume homogeneous feature spaces between source and target domains, which may not always be the case. Future research should focus on integrating different methods to leverage their combined strengths, exploring incremental complexity learning, and developing reward shaping techniques for complex tasks. Additionally, enhancing the realism of simulation environments and improving the robustness of transferred policies will be crucial for advancing this field. Other promising directions include exploring lifelong learning approaches, where models continuously learn and adapt to new tasks and environments, and developing more efficient algorithms for real-time adaptation and online learning.

VIII. Conclusion

Sim-to-real transfer is essential for deploying deep RL policies in real-world robotics. While domain randomization is widely used, other approaches such as policy distillation, meta-learning, and the use of world models and RNNs also show significant promise. Recent approaches include policy performance correlation, "green-screening" backgrounds, alternative 3d representations, and video generation. Future research should aim to address the existing challenges and further refine these methods to improve the effectiveness and efficiency of sim-to-real transfer techniques. By continuing to advance these strategies, we can unlock the full potential of robotic systems in diverse and dynamic real-world environments. The ongoing development of robust, efficient, and adaptable sim-to-real transfer techniques will play a critical role in the widespread adoption and success of robotic systems across various industries and applications.

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